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Disaster Relief Facility Network Design in Metropolises

Lu Zhen, Kai Wang, and Hu-Chen Liu

***Abstract*—This paper studies a strategic level decision problem for disaster relief facility network planning in metropolises. This problem is concerned with some long-term decisions on locating the emergency shelters, locating the supply and medical centers, and maintaining fleets of ambulances and transportation vehicles. These established facilities and deployed vehicles constitute a humanitarian logistic network for rapid responses and reliefs. An integer programming model is proposed to minimize the total cost of establishing such a disaster response network. A Lagrangian relaxation method is developed for solving the problem in large scales. To illustrate the proposed method's usage in practice, a demo example about its application in Shanghai is given. Some numerical experiments are also performed to validate the effectiveness and the efficiency of the proposed model and method.**

***Index Terms*—Emergency, integer programming, Lagrangian relaxation, location, relief operations.**

I. Introduction

THE provision for effective and cost efficient response and relief services is an important problem encountered in metropolises when facing some natural or man-made disasters. Nowadays, emergency planners tend to consider more operational level details in their decisions; but this cannot be accomplished without long-term commitments [1]. Preestablishment of emergency shelters, supply and medical centers, and some other resources is very important to enable a rapid and efficient response. By learning lessons from hurricane Katrina, the emergency management Department of U.S. established some warehoused, route planned shelters and material supply centers in hurricane prone regions. This problem deals with decisions such as: how many shelters and supply centers are needed and where should they be established in advance according to the anticipation of a disaster? How to plan such a humanitarian response network, in which residential areas are allocated to shelters and the shelters are also assigned to some certain supply centers? A good design of the disaster relief response network can increase the efficiency of people evacuation activities and also help the emergency management agency of a metropolis to coordinate the transport of commodities from supply centers to shelters in affected areas and the transport of wounded people from affected areas and shelters to medical units.

Manuscript received December 29, 2013; revised July 5, 2014; accepted September 25, 2014. This work was supported in part by the National Natural Science Foundation of China under Grant 71422007 and Grant 71101087, in part by the Program for Professor of Special Appointment (Eastern Scholar) at the Shanghai Institutions of Higher Learning, in part by the Excellent Young Faculty Research Program in Shanghai University under the 085 Project "Smart City and Metropolis Development," in part by the Shanghai Shu-Guang Project under Grant 11SG40, and in part by the Shanghai Pujiang Talent Program under Grant 11PJC072. This paper was recommended by Associate Editor K. W. Hipel. *(Corresponding author: Lu Zhen.)*

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Digital Object Identifier 10.1109/TSMC.2014.2364550

This paper develops an integer programming model for minimizing the total establishment cost of the disaster relief response network, which includes the decisions on locating the facilities and assignment relationships among them. For solving the model in large problem scales, a Lagrangian relaxation-based solution method is proposed. To illustrate the proposed method's usage in practice, a demo example about its application in Shanghai is given. Some numerical experiments are also performed to validate the effectiveness and the efficiency of the proposed model and method.

The remainder of this paper is organized as follows. Section II is the literature review. Problem background is elaborated in Section III. Then, the proposed model is given in Section IV. Sections V and VI state the Lagrangian relaxation-based solution method. Section VII illustrates a demo example of the model applied in Shanghai. Some numerical experiments are performed in Section VIII for a further investigation on the proposed model and method.

II. Related Works

There are a lot of studies on decision models to support evacuation and disaster response operations [2], [3]. These studies assume the shelter locations are known data. In some evacuation models, the location of shelters is variable. Sherali *et al.* [4] studied an evacuation planning problem, which tries to optimize the shelter locations and traffic flow distribution. However, it is not practical in reality because drivers are usually free to select their travelling routes [3]. Kongsomaksakul *et al.* [5] developed a bi-level shelter location model for investigating the impact of shelter locations on evacuation operations. A similar problem about shelter allocation was studied by Ng *et al.* [6]. Different from these studies, Kongsomaksakul *et al.* [5] formulated their model from a more operational level. Li *et al.* [7] developed a stochastic bi-level decision model for optimizing shelter locations. In their paper, the drivers' route choice behaviors were studied by dynamic equilibrium methods.

Besides the above studies on locating shelters, ambulance location and relocation problems also attract attentions from academia. Brotcorne *et al.* [8] investigated the ambulance location and assignment problems by defining three types of mathematical models. Jia *et al.* [9] proposed three deterministic models: 1) the first model is about how to cover demand points within a radius; 2) the second model is based on a P-median model and aims to minimize the total length of routes; and 3) the third model is based on a P-center model and aims to optimize the worst performance. Furthermore, Jia *et al.* [10] developed some heuristics for facility location of medical supplies in large-scale problem instances. Tzeng *et al.* [11] provided a multicriteria model, which can optimize the cost and the service level. Alsalloum and Rand [12] and Rajagopalan *et al.* [13] studied how to determine the minimum number of emergency vehicles and their locations. The former one used goal programming and the latter one considered dynamic ambulance relocation decisions. Yi and Özdamar [14] designed a location and routing network-flow model, which optimizes the coverage decisions so as to support evacuations. Uncertain factors are considered in some studies on disaster management, which used a methodology of stochastic programming to handle various uncertain factors [15]–[19]. Disaster management related problems were often formulated as two-stage decision models, in which the first stage is usually about the level of preparedness (e.g., location of facilities and inventory level of all types of resources) before a disaster happens, and the second stage is usually about how to response when uncertainty has been revealed [20]. This paper is mainly related to the above mentioned first stage. Snyder [21] discussed both advantages and disadvantages of two-stage models on disaster management.

Different from the above studies, this paper tries to optimize locations of emergency shelters and supply centers simultaneously. These two types of facilities not only affect rapid evacuation in the initial stage of disasters, but also involve long-term support for sustaining basic needs of survivors. This paper makes an explorative study on an integrated optimization problem for locating the two types of facilities.

This paper employed Lagrangian relaxation method to solve a problem in relief facility network. This method was also used for solving a decision model on logistics networks with expedited shipment services [22], and a decision model on reliable incapacitated fixed-charge distribution network [23]. Yu *et al.* [24] employed the Lagrangian relaxation for solving a configuration problem in the medical reference decision theoretic network. A near optimal diagnosis configuration is found by minimizing the duality gap via the subgradient method. Zhang *et al.* [25] also proposed a Lagrangian relaxation algorithm for solving a fault diagnosis problem of test selection in the presence of imperfect tests. Kodali *et al.* [26] studied a dynamic set-covering problem. By relaxing the coupling constraints using Lagrange multipliers, the problem was decoupled into independent subproblems. The Lagrange multipliers were updated using a subgradient method. Han *et al.* [27] used the Lagrangian relaxation method to solve a multilevel resource allocation model that is capable of allocating hierarchically organized assets to process interdependent tasks in order to accomplish mission objectives.

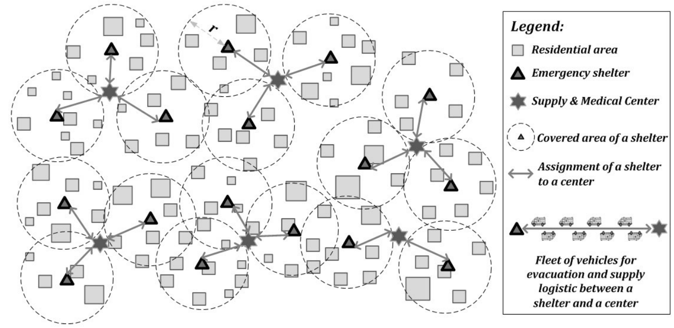

Fig. 1. Example of the disaster relief facility network.

Bernardi and Campos [28] used the Lagrangian relaxation method for analyzing a min-max model on computing lower bound of cycle time in interval-based time Petri nets. The Lagrangian relaxation methodologies were also widely used in port operation problems [29] and maritime shipping liner decision problems [30].

From the above, we can see that the Lagrangian relaxation method has been widely used for solving various types of problems and models including some decision models on network related problems. This paper also uses this method for solving a cost minimization model on disaster relief facility network.

III. Problem Backgrounds

When a natural or man-made disaster strikes a metropolis, evacuation and logistics support are two major activities in disaster response [14]. The first activity, i.e., evacuation of people from their residential areas to some near emergency shelters, takes place during the initial response phase; while the second activity, i.e., logistics support operations, tends to continue for a longer time for sustaining basic needs of survivors. Distance between residential areas and shelters, timely availability of commodities in shelters, and effective transportation of wounded people have significant influences on survival rate in the affected areas of the metropolis.

As shown in Fig. 1, a disaster relief facility network includes some emergency shelters, which should cover all the residential areas of the metropolis, and some supply and medical centers, which can deliver commodities to some dedicated shelters and also receive the wounded from the shelters. The establishment costs of shelters and supply and medical centers are the main parts of the total cost of building such a disaster relief facility network. Besides them, the cost of maintaining fleets of vehicles, which transport the wounded and commodities between the shelters and the centers, should also be considered. Given a pair of shelter and center, the cost of maintaining a fleet of vehicles between them can be estimated as known data in advance, which mainly depends on the distance between them.

IV. Model Formulation

In this section, an integer programming model is formulated for the above-mentioned problem. More specifically, all decision variables in the proposed model are binary variables, which reflect the structural features for the network of disaster relief facilities. For some nonlinear forms in constraints, they are also linearized by defining some auxiliary variables and constraints.

Before formulating the model, some assumptions in this paper are clarified as follows.

1) For each residential area, there is at least one emergency shelter that can cover the residential area.
2) Total population does not exceed total capacity of all possible emergency shelters, and also does not exceed total capacity of all supply and medical centers.

A. Index and Sets

- a Residential area locations.
- A Set of all residential area locations.
- s Emergency shelter locations.
- S Set of possible locations for establishing a new emergency shelter.
- S' Set of locations of existing emergency shelters.
- c Supply and medical center locations.
- C Set of possible locations for establishing a new supply and medical center.
- C' Set of locations of existing supply and medical centers.

B. Input Data

- p_a Population of the residential area a.
- d_{as} Distance between the residential area a and the emergency shelter s.
- r_s Radius of the area that can be covered by the emergency shelter s.
- m_s Maximum population that can be served by the emergency shelter s.
- n_c Maximum population that can be served by the supply and medical center c.
- e_s Cost of establishing a new emergency shelter in location s.
- f_c Cost of establishing a new supply and medical center in location c.
- t_{sc} Cost of maintaining a fleet of vehicles for transporting goods and people between the emergency shelter s and the supply and medical center c.

C. Decision Variables

- x_s Binary variable, equals to one if a new emergency shelter is established in location s, and zero otherwise; $s \in S$.
- y_c Binary variable, equals to one if a new supply and medical center is established in location c, and zero otherwise; $c \in C$.
- $\boldsymbol{g}_{as}$ Binary variable, equals to one if the residential area a is assigned to the emergency shelter s, and zero otherwise; $a \in A$, $s \in S \cup S'$.
- h_{sc} Binary variable, equals to one if a newly established emergency shelter s is assigned to the supply and medical center c, and zero otherwise; $s \in S$, $c \in C \cup C'$.

It should be noted that for the existing emergency shelters, i.e., $s \in S'$, their assignments to the supply and medical centers do not change during expansion of the emergency response network. Thus an input parameter h'_{sc}, which is similar with the above decision variable h_{sc}, is defined as follows.

- h'_{sc} Binary parameter, equals to one if an existing emergency shelter s is assigned to the supply and medical center c, and zero otherwise; $s \in S'$, $c \in C \cup C'$.

D. Mathematical Model

A mathematic model is formulated as follows:

$$(\boldsymbol{M\_0})\ \mathrm{Min} \sum_{s\in S} e_s x_s + \sum_{c\in C} f_c y_c + \sum_{s\in S, c\in C\cup C'} t_{sc} h_{sc} \tag{1}$$

$$s.t. \sum_{s\in S\cup S'} \boldsymbol{g}_{as} = 1 \qquad \forall a \in A \tag{2}$$

$$\sum_{c\in C\cup C'} h_{sc} = x_s \qquad \forall s \in S \tag{3}$$

$$\boldsymbol{g}_{as} \le x_s \qquad \forall a \in A,\ s \in S \tag{4}$$

$$h_{sc} \le y_c \qquad \forall s \in S, c \in C \tag{5}$$

$$d_{as}\boldsymbol{g}_{as} \le r_s \qquad \forall a \in A, s \in S \cup S' \tag{6}$$

$$\sum_{a\in A} p_a \boldsymbol{g}_{as} \le m_s \qquad \forall s \in S \cup S' \tag{7}$$

$$\sum_{a\in A} p_a \left(\sum_{s\in S} \boldsymbol{g}_{as} h_{sc} + \sum_{s\in S'} \boldsymbol{g}_{as} h'_{sc} \right) \le n_c \qquad \forall c \in C \cup C' \tag{8}$$

$$x_s \in \{0, 1\} \qquad \forall s \in S \tag{9}$$

$$y_c \in \{0, 1\} \qquad \forall c \in C \tag{10}$$

$$\boldsymbol{g}_{as} \in \{0, 1\} \qquad \forall a \in A, s \in S \cup S' \tag{11}$$

$$h_{sc} \in \{0, 1\} \qquad \forall s \in S, c \in C \cup C'. \tag{12}$$

Equation (2) state that each residential area is assigned to a shelter, and in the shelter's coverage. Equation (3) guarantee that each newly established shelter is assigned to a supply and medical center. Equation (4) ensure that residential areas cannot be assigned to a shelter location where no facility has been installed. Similarly, (5) ensures shelters cannot be assigned to a supply and medical center location where no facility has been installed. Equation (6) guarantees each residential area should be within the effective coverage of their assigned shelter. Equation (7) ensures that people assigned to each shelter cannot exceed its capacity. Similarly, (8) ensures that people served by each supply and medical center cannot exceed its capacity. Equations (9)–(12) define the decision variables.

E. Linearization of M_0

We need to linearize the nonlinear forms (i.e., $\delta_{as}\eta_{sc}$) in (8) of the above model M_0. An additional decision variable (z_{asc}) is defined to linearize it. Here z_{asc} can be regarded as the product of $\boldsymbol{g}_{as}$ and h_{sc} for $s \in S$ only. Then, (8) turns to

$$\sum_{a\in A} p_a \left(\sum_{s\in S} z_{asc} + \sum_{s\in S'} \boldsymbol{g}_{as} h'_{sc} \right) \le n_c \forall c \in C \cup C'. \tag{13}$$

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In order to reflect the product relationship between z_{asc}, $\boldsymbol{g}_{as}$, and h_{sc}, some additional constraints are needed as follows:

$$z_{asc} \le \boldsymbol{g}_{as} \qquad \forall a \in A, s \in S, c \in C \cup C' \tag{14}$$

$$z_{asc} \le h_{sc} \qquad \forall a \in A, s \in S, c \in C \cup C' \tag{15}$$

$$\boldsymbol{g}_{as} + h_{sc} \le z_{asc}+1 \qquad \forall a \in A, s \in S, c \in C \cup C' \tag{16}$$

$$z_{asc} \in \{0, 1\} \qquad \forall a \in A, s \in S, c \in C \cup C'. \tag{17}$$

Then the original model M_0 is linearized as follows:

$$(\mathbf{M\_0})\ (1)$$
$$s.t.\ (2)-(7);\ \ (9)-(17).$$

V. Lagrangian Relaxation of the Model

For the above proposed model M_0, some small-scale problem cases may be solved by the well-known branch and bound method. For some large-scale problem cases, the branch and bound method may be very time-consuming. This encourages us to develop a heuristic method for solving the proposed model M_0 in large-scale problem cases. The Lagrangian relaxation method is a widely-used heuristic for solving large-scale mixed integer programming models. Although the Lagrangian relaxation method belongs to heuristics, its key advantage is that it provides an approximate duality gap, which is an upper bound measure of suboptimality of the solution. The results obtained by the Lagrangian relaxation is usually more convincing than the results obtained by some other meta-heuristics such as simulated annealing, genetic algorithms. Moreover, because of the special structure of the proposed model M_0, the model after the Lagrangian relaxation can be divided into some independent models, which can also be solved in relatively easy manners. Therefore, this paper employs the Lagrangian relaxation methodology to develop a solution method for solving the proposed model M_0.

A. Lagrangian Relaxation Formulation

In this paper, the Lagrangian relaxation method is employed to reformulate the model (M_0) as the following model (M_LR), where (3), (4), (7), (13), (14), (16) are relaxed:

$$\begin{aligned}(\mathbf{M\_LR})\ Z_{LR}(\beta,\gamma,\delta,\eta,\theta,\lambda) &= \min \sum_{s\in S} e_s x_s + \sum_{c\in C} f_c y_c \\ &+ \sum_{s\in S, c\in C\cup C'} t_{sc} h_{sc} + \sum_{s\in S} \beta_s \left(\sum_{c\in C\cup C'} h_{sc} - x_s\right) \\ &+ \sum_{a\in A,\ s\in S} \gamma_{as}\left(\boldsymbol{g}_{as} - x_s\right) \\ &+ \sum_{s\in S\in S'} \delta_s \left(\sum_{a\in A} p_a \boldsymbol{g}_{as} - m_s\right) \\ &+ \sum_{c\in C\cup C'} \eta_c \left(\sum_{a\in A} p_a \left(\sum_{s\in S} z_{asc} + \sum_{s\in S'} \boldsymbol{g}_{as} h'_{sc}\right) - n_c\right) \\ &+ \sum_{a\in A, s\in S, c\in C\cup C'} \theta_{asc}\left(z_{asc} - \boldsymbol{g}_{as}\right) \\ &+ \sum_{a\in A, s\in S, c\in C\in C'} \lambda_{asc}\left(\boldsymbol{g}_{as} + h_{sc} - z_{asc} - 1\right) \end{aligned} \tag{18}$$

$$s.t. \sum_{s\in S\cup S'} \boldsymbol{g}_{as} = 1 \qquad \forall a \in A \tag{19}$$

$$h_{sc} \le y_c \qquad \forall s \in S, c \in C \tag{20}$$

$$d_{as}\boldsymbol{g}_{as} \le r_s \qquad \forall a \in A, s \in S \cup S' \tag{21}$$

$$z_{asc} \le h_{sc} \qquad \forall a \in A, s \in S, c \in C \cup C' \tag{22}$$

$$x_s \in \{0, 1\} \qquad \forall s \in S \tag{23}$$

$$y_c \in \{0, 1\} \qquad \forall c \in C \tag{24}$$

$$\boldsymbol{g}_{as} \in \{0, 1\} \qquad \forall a \in A, s \in S \cup S' \tag{25}$$

$$h_{sc} \in \{0, 1\} \qquad \forall s \in S, c \in C \cup C' \tag{26}$$

$$z_{asc} \in \{0, 1\} \qquad \forall a \in A, s \in S, c \in C \cup C'. \tag{27}$$

The reason for choosing the above six constraints for relaxation is that the above relaxed model can be decomposed into several independent models, which can be solved in a relatively easy manner. Moreover, in the above formulation, (14) are relaxed but (15) are not relaxed. If we consider (15) in (18), one more Lagrangian multiplier is needed. The efficiency of this formulation is lower than the above one. Theoretically, all the constraints can be relaxed in the objective by using the Lagrangian relaxation methodology. However, it will incur a low-efficient solving process. Thus we should find a good balance between the number of constraints and complexity of the objective.

B. Model Decomposition

Equation (18) in the above model (M_LR) can be reformulated as: $\pi(\beta,\gamma,\delta,\eta,\theta,\lambda) =$

$$\begin{aligned}&\text{Min} \sum_{s\in S}\left(e_s - \beta_s - \sum_{a\in A}\gamma_{as}\right)\mathbf{x_s} \\ &+ \sum_{a\in A}\left[\sum_{s\in S}\left(\gamma_{as} + \delta_s p_a + \sum_{c\in C\cup C'}(\lambda_{asc} - \theta_{asc})\right)\mathbf{g_{a}}_{s}\right. \\ &\qquad \left. + \sum_{s\in S'}\left(\delta_s p_a + \sum_{c\in C\cup C'} \eta_c p_a h'_{sc}\right)\mathbf{g_{a}}_{s}\right] \\ &+ \sum_{a\in A, s\in S, c\in C\cup C'} (\theta_{asc} + \eta_c p_a - \lambda_{asc})\,\mathbf{z_{a}}_{sc} \\ &+ \sum_{s\in S, c\in C\cup C'}\left(\sum_{a\in A}\lambda_{asc} + \beta_s + t_{sc}\right)\mathbf{h}_{sc} \\ &+ \sum_{c\in C} f_c \mathbf{y_c} - \sum_{a\in A, s\in S, c\in C\cup C'} \lambda_{asc} \\ &- \sum_{c\in C\cup C'} \eta_c n_c - \sum_{s\in S\cup S'} \delta_s m_s. \end{aligned} \tag{28}$$

The above objective is formulated according to the decision variables, which are in bold type. Then the model (M_LR) is decomposed into three independent models, which are denoted by M_LR_x, M_LR_g, and M_LR_{yhz}. The three models are based on the decision variables x_s, $\boldsymbol{g}_{as}$, and (y_c, h_{sc}, z_{asc}), respectively.

The reason for decomposing the model (M_LR) into the above-mentioned three models (M_LR_x, M_LR_g, M_LR_{yhz}) is mainly according to the structure of objective and constraints.

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In (28), the part with the variable $\mathbf{x_s}$ is independent from other variables, and there is no constraint on $\mathbf{x_s}$ (except for its definition constraints). Thus, it is proper to set an independent submodel on $\mathbf{x_s}$. The submodel is denoted by M_LR$_x$.

In (28), the part with the variable $\boldsymbol{g}_{as}$ is also independent from other variables. Equations (19) and (21) contain $\boldsymbol{g}_{as}$, but they do not connect with any other decision variables. Therefore, an independent submodel on $\boldsymbol{g}_{as}$ is formulated and denoted by M_LR$_g$.

Then, the remainder parts of objective, constraints, and variables are formulated as a submodel on the variables $\mathbf{y_c}$, $\mathbf{h_{sc}}$, and $\mathbf{z_{asc}}$. The submodel is denoted by M_LR$_{yhz}$

1) Model $\boldsymbol{M\_LR_x}$*:*

$$(\mathbf{M\_LR_x})\ \text{Min} \sum_{s\in S}\left(e_s-\beta_s-\sum_{a\in A}\gamma_{as}\right)x_s - \sum_{a\in A,s\in S,c\in C\cup C'}\lambda_{asc} - \sum_{c\in C\cup C'}\eta_c n_c - \sum_{s\in S\cup S'}\delta_s m_s \tag{29}$$

$$s.t.\quad x_s\in\{0,1\}\qquad \forall s\in S. \tag{30}$$

The optimal solution to the above model can be easily obtained according to

$$x_s=\begin{cases}1, & if\ \ e_s<\beta_s+\sum_{a\in A}\gamma_{as}\\ 0, & if\ \ e_s\geq\beta_s+\sum_{a\in A}\gamma_{as}\end{cases}\qquad \forall s\in S. \tag{31}$$

2) Model $\boldsymbol{M\_LR_g}$*:*

$$(M\_LR_{\mathbf{g}})\ \text{Min}\sum_{a\in A}\left[\sum_{s\in S}\left(\gamma_{as}+\delta_s p_a + \sum_{c\in C\cup C'}(\lambda_{asc}-\theta_{asc})\right)\boldsymbol{g}_{as} + \sum_{s\in S'}\left(\delta_s p_a+\sum_{c\in C\cup C'}\eta_c p_a h'_{sc}\right)\boldsymbol{g}_{as}\right] \tag{32}$$

$$s.t.\quad \sum_{s\in S\cup S'}\boldsymbol{g}_{as}=1\quad \forall a\in A \tag{33}$$

$$d_{as}\boldsymbol{g}_{as}\leq r_s\qquad \forall a\in A, s\in S\cup S' \tag{34}$$

$$\boldsymbol{g}_{as}\in\{0,1\}\qquad \forall a\in A, s\in S\cup S'. \tag{35}$$

The above model (M_LR$_g$) can be decomposed into $|A|$ independent subproblems. Given a value of a, only one $\boldsymbol{g}_{as}$ equals to one for all $s\in S\cup S'$, and others are set to zero. The following procedure in Table I can determine which $\boldsymbol{g}_{as}$ should be set as one for all $s\in S\cup S'$. The main idea is to calculate the two types of coefficients for $\boldsymbol{g}_{as}$. Their values are denoted by temp$_{as}$ in the procedure. The $\boldsymbol{g}_{as}$ variable whose coefficient reaches the minimum value will be set as one, and other $\boldsymbol{g}_{as}$ variables will be set as zero.

3) Model $\boldsymbol{M\_LR_{yhz}}$*:*

$$(\mathrm{M\_LR}_{\mathbf{y}hz})\ \min\sum_{a\in A,s\in S,c\in C\cup C'}(\theta_{asc}+\eta_c p_a-\lambda_{asc})z_{asc} + \sum_{s\in S,c\in C\cup C'}\left(\sum_{a\in A}\lambda_{asc}+\beta_s+t_{sc}\right)h_{sc}+\sum_{c\in C}f_c y_c \tag{36}$$

$$s.t.\ h_{sc}\leq y_c\qquad \forall s\in S, c\in C \tag{37}$$

$$z_{asc}\leq h_{sc}\qquad \forall a\in A, s\in S, c\in C\cup C' \tag{38}$$

$$y_c\in\{0,1\}\qquad \forall c\in C \tag{39}$$

$$h_{sc}\in\{0,1\}\qquad \forall s\in S, c\in C\cup C' \tag{40}$$

$$z_{asc}\in\{0,1\}\qquad \forall a\in A, s\in S, c\in C\cup C'. \tag{41}$$

TABLE I
PROCEDURE FOR SOLVING M_LR$_g$

```
For all a ∈ A
    For all s ∈ S ∪ S'
        g_as* ← 0
        If d_as ≤ r_s Then
            If s ∈ S Then
                temp_as ← γ_as + δ_s p_a + Σ_{c∈C∪C'} (λ_asc − θ_asc)
            Else
                temp_as ← δ_s p_a + Σ_{c∈C∪C'} η_c p_a h'_sc
            End If
        End If
    End For
    s* ← arg min_{s∈S∪S'} temp_as
    g_as* ← 1
End For
```

As all the decision variables have a common subscript c, $c\in C\cup C'$. Thus the above model (M_LR$_{yhz}$) can be decomposed into $|C\cup C'|$ independent subproblems, which can be solved easily. The items in the previous objective [see (36)] are divided into two categories. One is about the set C, and the other is about the set C'. In this way, the objective of the model is reformulated as follows:

$$\min\sum_{c\in C}\left\{\sum_{s\in S}\left[\left(\sum_{a\in A}\lambda_{asc}+\beta_s+t_{sc}\right)h_{sc} + \sum_{a\in A}(\theta_{asc}+\eta_c p_a-\lambda_{asc})z_{asc}\right]+f_c y_c\right\} + \sum_{c\in C'}\left\{\sum_{s\in S}\left[\left(\sum_{a\in A}\lambda_{asc}+\beta_s+t_{sc}\right)h_{sc} + \sum_{a\in A}(\theta_{asc}+\eta_c p_a-\lambda_{asc})z_{asc}\right]\right\}. \tag{42}$$

The following procedure shown in Table II, can be used for solving the above model (M_LR$_{yhz}$) by some decomposing methods. The procedure contains three-layer loops that determine the variables (z_{asc}, h_{sc}, y_c) along the iteration order of the inner-layer, middle-layer, and outer-layer, respectively. These variables are mainly determined according to their coefficients. Moreover, in the procedure, "temp1" denotes the value of "$\sum_{s\in S}[(\sum_{a\in A}\lambda_{asc}+\beta_s+t_{sc})h_{sc}+\sum_{a\in A}(\theta_{asc}+\eta_c p_a-\lambda_{asc})z_{asc}]$," and "temp2" denotes the value of "$\theta_{asc}+\eta_c p_a-\lambda_{asc}$." The temp1 and temp2 are also used to determine the values of binary variables (z_{asc}, h_{sc}, y_c).

VI. SOLUTION PROCEDURE

The subgradient method is an iterative method for solving convex minimization problems. Some other methods, e.g., interior-point methods, bundle methods, have also been suggested for solving convex minimization problems. However, the subgradient method still remains competitive for some situations. Moreover, the subgradient method was widely used in plenty of studies that used the Lagrangian relaxation. Therefore, this paper still uses the subgradient method for solving the Lagrangian relaxation model as the usual practice.

A. Subgradient Method

The solution obtained by the above Lagrangian relaxation model (M_LR) depends on the Lagrangian multipliers β, γ, δ, η, θ, and λ. If $Z_{LR}(\beta,\gamma,\delta,\eta,\theta,\lambda)$ is the value of the Lagrangian function with the multiplier sets β, γ, δ, η, θ, and

TABLE II
PROCEDURE FOR SOLVING M_LR$_{yhz}$

```
For all c ∈ C ∪ C′
    temp1 ← 0
    For all s ∈ S
        temp2 ← 0
        For all a ∈ A
            If θ_asc + η_c p_a − λ_asc < 0 Then
                z_asc ← 1
                temp2 ← temp2 + θ_asc + η_c p_a − λ_asc
            Else
                z_asc ← 0
            End If
        End For
        If ∑_{a∈A} λ_asc + β_s + t_sc + temp2 < 0 Then
            h_sc ← 1
            temp1 ← temp1 + ∑_{a∈A} λ_asc + β_s + t_sc + temp2
        Else
            h_sc ← 0; z_asc ← 0 for all a ∈ A
        End If
    End For
    If c ∈ C Then
        If f_c + temp1 < 0 Then
            y_c ← 1
        Else
            h_sc ← 0 for all s ∈ S; z_asc ← 0 for all a ∈ A, s ∈ S
            y_c ← 0
        End If
    End If
End For
```

TABLE III
PROCEDURE FOR THE SUBGRADIENT METHOD

```
Z_BestF ← +∞, Z_BestLB ← 0, (β,γ,δ,η,θ,λ) ← {0,0,0,0,0,0}, π^1 ← 2,
mark← 1
For k from 1 to 200      // 200 is the maximum number of iterations
    Solve M_LR; its objective solution value is denoted by Z_LR
    If Z_LR > Z_BestLB Then
        Z_BestLB ← Z_LR, mark← 1, (β*,γ*,δ*,η*,θ*,λ*) ← (β,γ,δ,η,θ,λ)
    Else
        mark← mark + 1
    End If
    Obtain a feasible solution by a heuristic, its objective solution value is
    denoted by Z_Feas
    If Z_Feas < Z_BestF Then
        Z_BestF ← Z_Feas, mark← 1, (β*,γ*,δ*,η*,θ*,λ*) ← (β,γ,δ,η,θ,λ)
    End If
    If Z_BestF = Z_BestLB Then
        Stop the whole procedure
    End If
    If mark > 20 Then
        mark← 1, (β,γ,δ,η,θ,λ) ← (β*,γ*,δ*,η*,θ*,λ*), π^k ← π^k/2
    End If
    Update multipliers (β,γ,δ,η,θ,λ) according to formulae (44-55)
    If some of the multipliers (β,γ,δ,η,θ,λ) < 0 Then
        Set them to zero
    End If
End For
```

λ, the best lower bound corresponding to the optimal multiplier sets β^*, γ^*, δ^*, η^*, θ^*, and λ^* is determined as

$$Z_{LR}\left(\beta^*, \gamma^*, \delta^*, \eta^*, \theta^*, \lambda^*\right) = \mathrm{Max}_{\beta,\gamma,\delta,\eta,\theta,\lambda}\left(Z_{LR}\left(\beta, \gamma, \delta, \eta, \theta, \lambda\right)\right). \quad (43)$$

The widely used subgradient optimization procedure is applied in this paper to find the sets of the Lagrangian multipliers. During each iteration of the procedure, M_LR is solved to obtain a lower bound for M_0. The solution obtained by M_LR may not be feasible for M_0, thus a procedure is performed at each iteration for determining a feasible solution of M_0. At the time of termination, the subgradient optimization procedure reports the best feasible solution and the best lower bound, which are generated in all the iterations.

In the subgradient procedure, given a set of starting multipliers β^0, γ^0, δ^0, η^0, θ^0, and λ^0, the sequences of multipliers in iterations are generated using the following formulas:

$$\beta_s^{k+1} = \beta_s^k + \triangle\beta^k\left(\sum_{c\in C\cup C'} h_{sc} - x_s\right) \quad \forall s \in S \quad (44)$$

$$\gamma_{as}^{k+1} = \gamma_{as}^k + \triangle\gamma^k\left(\boldsymbol{g}_{as} - x_s\right) \quad \forall a \in A,\ s \in S \quad (45)$$

$$\delta_s^{k+1} = \delta_s^k + \triangle\delta^k\left(\sum_{a\in A} p_a \boldsymbol{g}_{as} - m_s\right) \quad \forall s \in S \cup S' \quad (46)$$

$$\eta_c^{k+1} = \eta_c^k + \triangle\eta^k\left(\sum_{a\in A} p_a\left(\sum_{s\in S} z_{asc} + \sum_{s\in S'} \boldsymbol{g}_{as} h'_{sc}\right) - n_c\right) \quad \forall c \in C \cup C' \quad (47)$$

$$\theta_{asc}^{k+1} = \theta_{asc}^k + \triangle\theta^k\left(z_{asc} - \boldsymbol{g}_{as}\right) \quad \forall a \in A, s \in S, c \in C \cup C' \quad (48)$$

$$\lambda_{asc}^{k+1} = \lambda_{asc}^k + \triangle\lambda^k\left(\boldsymbol{g}_{as} + h_{sc} - z_{asc} - 1\right) \quad \forall a \in A, s \in S, c \in C \cup C' \quad (49)$$

where $\triangle\beta^k$, $\triangle\gamma^k$, $\triangle\delta^k$, $\triangle\eta^k$, $\triangle\theta^k$, and $\triangle\lambda^k$ are some positive scalar step sizes in the kth iteration. They are determined by the following formulas:

$$\triangle\beta^k = \pi^k\left(Z_{\mathrm{best}F} - Z_{LR}(\beta, \gamma, \delta, \eta, \theta, \lambda)^k\right) / \left\|\sum_{c\in C\cup C'} h_{sc} - x_s\right\| \quad (50)$$

$$\triangle\gamma^k = \pi^k\left(Z_{\mathrm{best}F} - Z_{LR}(\beta, \gamma, \delta, \eta, \theta, \lambda)^k\right) / \left\|\boldsymbol{g}_{as} - x_s\right\| \quad (51)$$

$$\triangle\delta^k = \pi^k\left(Z_{\mathrm{best}F} - Z_{LR}(\beta, \gamma, \delta, \eta, \theta, \lambda)^k\right) / \left\|\sum_{a\in A} p_a \boldsymbol{g}_{as} - m_s\right\| \quad (52)$$

$$\triangle\eta^k = \pi^k\left(Z_{\mathrm{best}F} - Z_{LR}(\beta, \gamma, \delta, \eta, \theta, \lambda)^k\right) / \left\|\sum_{a\in A} p_a\left(\sum_{s\in S} z_{asc} + \sum_{s\in S'} \boldsymbol{g}_{as} h'_{sc}\right) - n_c\right\| \quad (53)$$

$$\triangle\theta^k = \pi^k\left(Z_{\mathrm{best}F} - Z_{LR}(\beta, \gamma, \delta, \eta, \theta, \lambda)^k\right) / \left\|z_{asc} - \boldsymbol{g}_{as}\right\| \quad (54)$$

$$\triangle\lambda^k = \pi^k\left(Z_{\mathrm{best}F} - Z_{LR}(\beta, \gamma, \delta, \eta, \theta, \lambda)^k\right) / \left\|\boldsymbol{g}_{as} + h_{sc} - z_{asc} - 1\right\| \quad (55)$$

where $Z_{\mathrm{best}F}$ is the best known feasible solution value; $\|\ldots\|$ denotes the Euclidean norm; π^k is a scalar satisfying that $0 < \pi^k \leq 2$. π^k is set to two at the beginning of the procedure and is halved whenever the lower bound does not improve in twenty consecutive iterations. The whole procedure of the subgradient method is shown in Table III. In the procedure, we define a parameter "mark" to denote the number of consecutive iterations in which the lower bound has not been improved. The core idea of the procedure is to change the Lagrangian multipliers along iterations. By using the defined parameter mark, the scalar π^k is halved when the lower bound does not improve in twenty consecutive iterations. Then, based on

TABLE IV
HEURISTIC FOR OBTAINING A FEASIBLE SOLUTION OF ORIGINAL MODEL (M_0)

For all $s \in S \cup S'$
  If $\sum_{a\in A} p_a g_{as} > m_s$ **Then**
    Adjust some g_{as} to disconnect the farthest residential area until $\sum_{a\in A} p_a g_{as} \le m_s$
  End If
  For the above disconnected residential area
    Assign them to another existing emergency shelter with extra capacity and close enough to cover them
    If no such emergency shelters available **Then**
      Set up a closest new emergency shelter
    End If
  End For
End For
For all $s \in S'$
  Assign the emergency shelter s to the nearest supply and medical center
End For
For all $c \in C \cup C'$
  If $\sum_{a\in A} p_a (\sum_{s\in S} g_{as} h_{sc} + \sum_{s\in S'} g_{as} h'_{sc}) > n_c$ **Then**
    Adjust some h_{sc} to disconnect the farthest emergency shelter until $\sum_{a\in A} p_a (\sum_{s\in S} g_{as} h_{sc} + \sum_{s\in S'} g_{as} h'_{sc}) \le n_c$
  End If
  For the above disconnected residential areas
    Assign them to another existing supply and medical center that has extra capacity and is close to them
    If no such supply and medical centers available **Then**
      Set up a closest new supply and medical center
    End If
  End For
End For

the scalar π^k, all the six types of Lagrangian multipliers are updated for the next iteration.

B. Heuristic for Obtaining Feasible Solution

As M_LR relaxes (3), (4), (7), (13), (14), (16), the solution of M_LR may not be feasible for the original model (M_0). A heuristic is proposed to obtain a feasible solution for M_0 based on the solution for M_LR, which is solved by the above subgradient method. The heuristic is shown in Table IV as follows. The core idea of the heuristic is to disconnect residential areas that are far away from a shelter if the shelter's capacity is exceeded. Then these disconnected residential areas are assigned to some near shelter with extra capacity; if none available shelter can be found, a new shelter will be set up. The newly established shelters will be assigned to a near supply and medical center accordingly.

VII. Demo Example in Shanghai

This section gives a demo example for application of the proposed model in Shanghai, which is the largest city by population in China. Shanghai has a population of over 23 million and a land area of about 6340 km^2. In such a megalopolis, there are about 500 residential area locations, i.e., $|A| = 500$; the population of a residential area (i.e., p_a) is between 1×10^4 and 3×10^4 approximately; there already exist ten emergency shelters and three supply centers, i.e., $|S'| = 10$, and $|C'| = 3$; there are 300 possible locations for establishing a new emergency shelter, and 60 possible locations for establishing a new supply center, i.e., $|S| = 300$, and $|C| = 60$. For radius of the area that can be covered by an emergency shelter, i.e., r_s, its average value is between 11 and 15 km. For the supply and medical centers, their capacities depend on their scales. The maximum population that can be served by a center (i.e., n_c) is between 3×10^5 and 9×10^5.

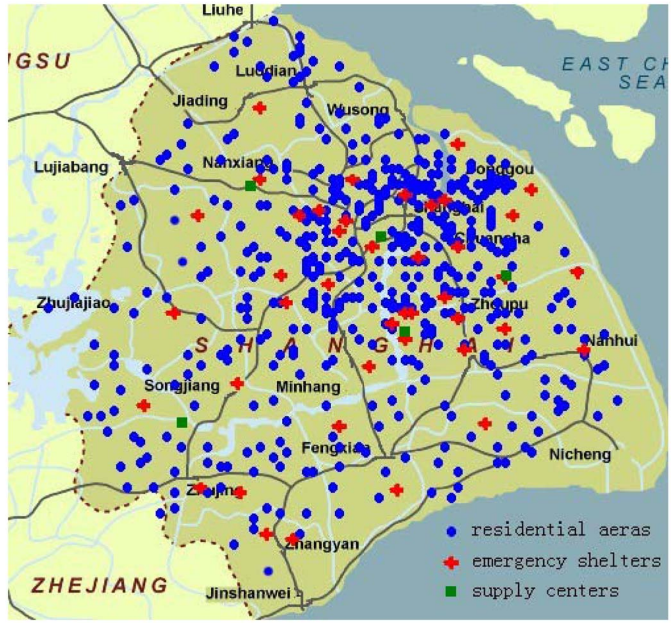

Fig. 2. Disaster relief facility network of Shanghai.

By using the proposed model and the Lagrangian relaxation-based solution method, the disaster relief facility network for Shanghai is obtained and shown in Fig. 2. Two supply centers and 34 shelters should be newly established. The total cost is about 39 167 810 RMB. For the demo example, the Lagrangian lower bound is 37 263 445, and the gap is about 4.9%.

VIII. Numerical Experiments

The experiments in this paper include three parts: 1) the comparison between the Lagrange relaxation-based solution method and the CPLEX; 2) the comparison between the proposed model and other intuitive method; and 3) numerical investigation on the parameter analysis.

A. Comparison Between the CPLEX and the Lagrange Relation-Based Method

We conduct some experiments to compare the results obtained by the Lagrange relaxation-based solution method and the optimal objective value solved by the CPLEX directly. The mathematical model is implemented by CPLEX12.1 with concert technology of C# (VS2008) on a PC with 1.6 GHz quad-core processor and 4 GB memory. The demo example in Shanghai is a large-scale case, in which ($|A|$,$|S|$,$|S'|$,$|C|$,$|C'|$) = (500, 300, 10, 60, 3). Here, the values of "$|A|$,$|S|$,$|S'|$,$|C|$,$|C'|$" reflect the scale of problem cases. All the instances used in this paper are generated on the basis of the baseline data set for the demo example in Shanghai.

Table V shows the results of comparative experiments on 12 small-scale cases. From the results in Table V, we can see that the CPLEX can only solve some very small

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TABLE V
COMPARISON BETWEEN CPLEX AND THE LAGRANGE RELAXATION-BASED METHOD

Cases						Lagrange Relaxation				CPLEX		GAP	First Feasible Solution	
#	$\|A\|$	$\|S\|$	$\|S'\|$	$\|C\|$	$\|C'\|$	OBJ_L	T	LB	GAP_{LB}	OBJ_C	T		CPLEX	Lagrange
1-1	70	20	10	4	1	14859584	39	14232460	4.2%	14516510	156	2.4%	22335487	32248154
1-2	70	25	10	4	1	14265465	44	13580683	4.8%	14055141	276	1.5%	19028365	29832656
1-3	70	30	10	4	1	13522583	49	12887478	4.7%	13156156	697	2.8%	18524473	27263593
1-4	70	30	12	4	1	13256211	58	12473462	5.9%	heuristic still looking		N.A.	N.A.	26808769
1-5	60	20	10	4	1	14325051	32	13780757	3.8%	13815471	76	3.7%	19252360	24828567
1-6	60	25	10	4	1	13456148	35	12497965	7.1%	13051515	184	3.1%	17823471	22740235
1-7	60	30	10	4	1	12846544	36	12035822	6.3%	12546120	334	2.4%	18624724	24936901
1-8	60	30	12	4	1	12548647	39	11883856	5.3%	heuristic still looking		N.A.	N.A.	25824790
1-9	50	20	10	4	1	14024547	20	13365491	4.7%	13563546	45	3.4%	20363586	25481526
1-10	50	25	10	4	1	13265106	24	12364953	6.8%	12881432	104	3.0%	19254279	23244478
1-11	50	30	10	4	1	12548562	30	11797537	6.0%	12323955	209	1.8%	18826492	22166743
1-12	50	30	12	4	1	12548562	36	12136779	3.3%	heuristic still looking		N.A.	N.A.	23219472
										Avg.:		**2.7%**		

Note: When solving the cases '1-4' '1-8' '1-12', the CPLEX solver halts after a short time, and shows 'heuristic still looking…' on the screen for a relatively long time. It means the solving process is extremely time-consuming, and the cases' scales are too large for the CPLEX solver to handle within a reasonable time. 'GAP' means the gap between OBJ_L & OBJ_C. 'T' means the CPU time, the unit is second.

problem cases. The Lagrange relaxation-based method can obtain near optimal solutions with average gap at about 2.7%. For the computation time of the two ways, the Lagrange relaxation-based method can solve the problem cases much faster than the CPLEX. According to the above results, it can be validated that the proposed Lagrange relaxation-based solution method can solve the original model M0 in an efficient way.

B. Comparison Between the Proposed Model and Other Method

To validate the effectiveness of the proposed mathematical model (M0), some experiments are performed for comparing it with another intuitive planning method. The intuitive method is described as follows.

1) Sequence all the residential areas by the decreasing order of their populations. Assign emergency shelters and supply centers to the residential areas one by one according to the above sequence.
2) For each residential area in the sequence, select a shelter from the set of available shelters and assign it to the residential area, which should be covered by the shelter. In addition, the assignment decision should not violate the capacity constraint of the shelter. If all the available shelters cannot satisfy the requirements, a new shelter is needed. The criterion for selecting a shelter location (s) from S, i.e., the set of possible locations for establishing a new shelter, mainly depends on its cost and capacity. More specifically, the selected shelter location is determined by: $s^* = \arg\mathrm{Min}_{s\in S}\{(e_s/\mathrm{Avg}_{j\in S}\, e_j) - (m_s/\mathrm{Avg}_{j\in S}\, m_j)\}$. Here, e_s is the cost of setting a new shelter in location s; m_s is the maximum population that can be served by the shelter s.
3) Similar with 2), assign a supply center or establish a new center to each shelter. The criterion for selecting a supply center (c) from C, i.e., the set of possible locations for establishing a new center, mainly depends on its cost and capacity. The selected shelter location is determined by: $c^* = \arg\mathrm{Min}_{c\in C}\{(f_c/\mathrm{Avg}_{j\in C}\, f_j) - (n_c/\mathrm{Avg}_{j\in C}\, n_j)\}$. Here, f_c is the cost of establishing a new center in location c; n_c is the maximum population that can be served by the center c.

TABLE VI
COMPARISON BETWEEN THE PROPOSED MODEL AND AN INTUITIVE METHOD

Cases						Intuitive method	Proposed model	OBJ
#	$\|A\|$	$\|S\|$	$\|S'\|$	$\|C\|$	$\|C$	OBJ	OBJ	Gap
1	600	300	60	10	3	50654845	46161512	9%
2	600	300	80	10	3	48151616	42640944	13%
3	600	300	100	10	3	46281211	41651601	11%
4	600	350	100	10	3	44125161	39414541	12%
5	600	400	100	10	3	43266514	37064848	17%
6	500	300	60	10	3	45899860	39167810	17%
7	500	300	80	10	3	44244940	38169048	16%
8	500	300	100	10	3	42635625	35399712	20%
9	500	350	100	10	3	41786786	34063040	23%
10	500	400	100	10	3	40323208	33346716	21%
11	400	300	60	10	3	42560489	35840792	19%
12	400	300	80	10	3	41604880	33519026	24%
13	400	300	100	10	3	41065068	31034564	32%
14	400	350	100	10	3	39645054	29568011	34%
15	400	400	100	10	3	37940145	28644840	33%
							Avg.	20%

Some experiments are conducted for 15 large-scale problem cases. The proposed model is solved by the Lagrange relaxation-based method. The comparative results by the proposed model and the intuitive method are listed in Table VI.

Table VI shows the model outperforms the intuitive method evidently. In these numerical cases, the average improvement rate for the proposed model is about 20% with respect to the objective value, i.e., the total cost. The results in Table VI validate the effectiveness of the proposed model in this paper. The model can act as a decision support tool for the disaster relief facility network planning in metropolises.

C. Parameter Analysis on the Disaster Relief Facility Network Planning

Experiments are performed to investigate the influences of parameters on the total establishment cost of facility network.

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TABLE VII
INFLUENCE OF SHELTERS' COVERAGE RADIUS ON THE TOTAL COST

Cases #	Average radius of shelters' coverage	OBJ (Total cost)
3-1	10 km	63679160
3-2	11 km	53216708
3-3	12 km	45960216
3-4	13 km	39167810
3-5	14 km	33083118
3-6	15 km	27910124
3-7	16 km	24974160
3-8	17 km	23124932
3-9	18 km	23124932
3-10	19 km	23124932

TABLE VIII
INFLUENCE OF RESIDENTIAL POPULATION DISTRIBUTION ON THE TOTAL COST

Cases #	Residential areas' population		OBJ
	Average	*Standard deviation*	(Total cost)
4-1	20207	2364	37541312
4-2	20259	2954	37591008
4-3	20311	3545	37472896
4-4	20363	4136	39697688
4-5	20415	4727	39108248
4-6	20467	5318	39188464
4-7	20519	5909	39167810
4-8	20571	6500	38448080
4-9	20623	7091	38510184
4-10	20674	7682	38592304

TABLE IX
INFLUENCE OF SHELTER COST ON THE TOTAL COST

Cases #	Increasing rate on the demo example		OBJ
	Shelter capacity	*Shelter cost*	(Total cost)
5-1	5%	1%	39289304
5-2		2%	39410824
5-3		3%	39532348
5-4		4%	39653868
5-5		5%	39775404
5-6		6%	39896912
5-7		7%	40018440
5-8		8%	40139962
5-9		9%	40261485
5-10		10%	40383008

TABLE X
INFLUENCE OF SHELTER CAPACITY ON THE TOTAL COST

Cases #	Increasing rate on the demo example		OBJ
	Shelter capacity	*Shelter cost*	(Total cost)
5-1	15%	5%	39775404
5-2	20%		39775404
5-3	25%		39775404
5-4	30%		39707876
5-5	35%		39707876
5-6	40%		39707876
5-7	45%		39307108
5-8	50%		39307108
5-9	55%		39307108
5-10	60%		39307108

1) Influence of Shelters' Coverage Radius: Experiments on ten problem cases are performed under different settings with respect to the average radius of shelters' coverage areas. Other parameter settings for the ten cases are the same as the demo example in Shanghai, i.e., $(|A|, |S|, |S'|, |C|, |C'|) = (500, 300, 10, 60, 3)$. The experimental results are listed in Table VII. From the results, it is observed that the cost decreases obviously with the coverage radius increasing. However, when the average radius exceeds 17 km, the cost evolves to a stable value, which implies the increasing shelter coverage radius does not always reduce the total cost.

2) Influence of Residential Population Distribution: Experiments on ten problem cases are performed to investigate whether the residential population distribution has influence on the total cost. More specifically, we analyze the relationship between the standard deviation of all the residential areas' populations and the total cost. In these experiments, other parameter settings for the ten cases are the same as the demo example in Shanghai. The experimental results are listed in Table VIII.

In Table VIII, the third column denotes the standard deviations of residential areas' population. The larger is the standard deviation, the population of the city is distributed in a more uneven manner. From the data in the third column, it is noted that the standard deviations increase evidently, but the objective values do not increase significantly. This result shows that whether a city's population is distributed evenly or unevenly has no obvious influence on the total cost of establishing the disaster relief facility network for the city.

3) Influence of Facility Cost and Capacity: The disaster relief facilities are mainly emergency shelters and supply centers. For the shelters, experiments on 30 problem cases are performed to investigate the influence of shelters' average cost and capacity on the total cost.

In the parameter settings of the following data tables, we use "increasing rate" to reflect the changing of parameters in the problem. The increasing rate is based on the baseline dataset of the demo example in Shanghai. For example, if the shelter capacity in the demo example is 100, the increasing rate "5%" means the capacity is 105 in the problem case. For the shelter cost, the meaning of increasing rate is similar.

Table IX shows the results of experiments, in which shelter capacity increases by 5% on the basis of the demo example, but the average cost of shelters increases by different degrees. The results in Table IX show that the increasing of the unit cost of shelters will incur the increasing of the total cost.

On the contrary, the experiments in Table X are performed in a different way from Table IX. In the experiments, the average cost of shelters increases by 5% on the basis of the demo example, but the shelter capacity increases by different degrees. The results in Table X imply that the increasing of the shelters' average capacity cannot reduce the total cost evidently.

The results in Tables IX and X show that based on the current situation of the disaster relief facility network in Shanghai, increasing the shelter capacity is not important for controlling the total cost of the network, but the average shelter cost still has a significant influence on the total cost.

TABLE XI
INFLUENCE OF SHELTER COST AND CAPACITY ON THE TOTAL COST

Cases #	Increasing rate on the demo example		OBJ
	Shelter capacity	*Shelter cost*	(Total cost)
5-1	15%	1%	39289304
5-2	20%	2%	39410824
5-3	25%	3%	39131580
5-4	30%	4%	39586340
5-5	35%	5%	39707876
5-6	40%	6%	39829384
5-7	45%	7%	36124668
5-8	50%	8%	36234640
5-9	55%	9%	36344608
5-10	60%	10%	36454592

Table XI shows the experiment results by changing the increasing rate of shelter average capacity and average cost at the same time. It is observed that the increasing of the shelters' average cost does not always incur the increasing of the total cost if the shelter average capacity also grows up simultaneously. When the shelter average capacity exceeds some certain level, the total cost for establishing the disaster relief facility network will drop down by an evident degree. The reason for this phenomenon lies in that fewer shelters need to be established if their average capacity exceeds a threshold value; then it results in an evident decrease of the total cost for establishing the disaster relief facility network in a metropolis.

Table XI is the experiments on the influence of shelter cost and capacity. The results on the influence of supply center cost and capacity are similar. Thus the results are omitted here.

IX. CONCLUSION

This paper studies a strategic level decision problem for disaster relief facility network planning in metropolises so as to realize an effective and cost efficient response and relief services in the metropolises when facing some natural or man-made disasters. An integer programming model is proposed and a solution method based on Lagrange relaxation is also developed to solve the model. A demo example of its application in Shanghai is illustrated. Some numerical experiments are also conducted for further investigations on the facility network planning decisions. The effective of the proposed model and the efficiency of the solution method are validated by the numerical experiments.

There are also limitations in this paper. The current model does not consider the dynamic changing of population distribution along the time. In addition, the radius of shelters and connecting costs between shelters and supply centers may not be deterministic but depend on the spatial patterns of the traffic situations, which are also dynamic changing along the time. All of these factors will further complicate the current model. These issues will be considered in our studies in the future.

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